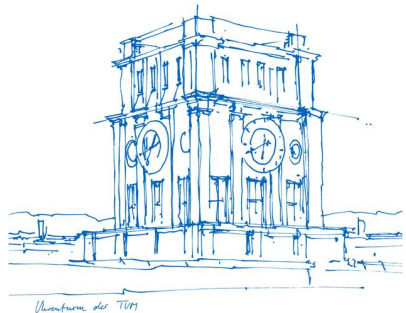


PoreAccess-CO₂: From Random Packing to an Interpretable Transport Closure

Structure-resolved diffusion and adsorption in
DEM-generated zeolite beds

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CO₂ adsorption matters and packing structure is a missing scale

Why CO₂ capturing matters?

- Food systems contribute about **one-third** of global anthropogenic greenhouse-gas emissions.
- Food and bioprocessing include recoverable biogenic CO₂ streams, for example from fermentation.
- Regenerable sorbents offer a compact capture and purification route.

What is already established?

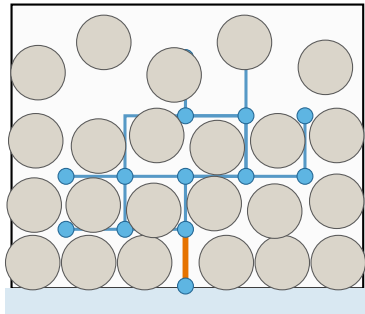
- CO₂ equilibrium and uptake kinetics on zeolite 13X are widely studied.
- Fixed-bed models normally use bulk porosity and a fitted effective transport coefficient.
- Random inlet-connected paths remain hidden inside that coefficient.

Novelty: connect random particle geometry to a fast bed-scale uptake model through a physically calculated pore-accessibility descriptor.

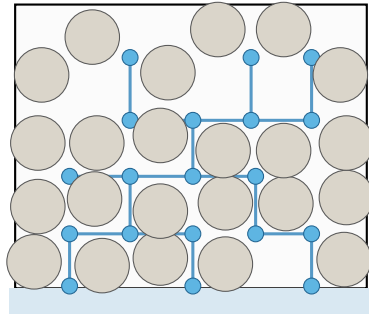
Same porosity can hide different inlet-connected pathways

Can a pore-accessibility measure predict bed-scale transient transport better than porosity alone?

Packing A: inlet bottleneck



Packing B: multiple inlet paths



- pore body
- diffusive throat
- zeolite bead
- inlet bottleneck

Same beads and similar porosity; different numbers and conductances of inlet-connected paths.

Why zeolite 13X and what does the study isolate?

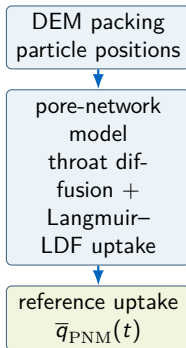
Why zeolite 13X?

- established benchmark sorbent for CO₂
- strong equilibrium uptake at low partial pressure
- commercial millimetre-scale beads form random beds

Controlled model problem

- dry, isothermal, stagnant gas
- CO₂ enters from a reservoir below the bed
- pore-network model resolves throat diffusion and bead adsorption
- no imposed flow; no CFD

How strongly does random void connectivity change access to the adsorbent?



Twenty beds isolate variability caused by random arrangement

Quantity	Fixed value
beads per bed	1850
bead diameter	1.6 mm
tube diameter	16 mm
D_{tube}/d_p	10
temperature	298.15 K
inlet c_{CO_2}	6.05 mol m^{-3}
simulation time	5000 s

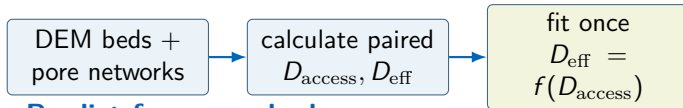
Only the DEM insertion seed changed

- Delaunay–Voronoi networks were built directly from particle coordinates
- all networks were connected and spanning
- quasi-uniform Sobol sampling allocated pore volumes without a voxel grid
- transient and 1D models passed mass-balance checks

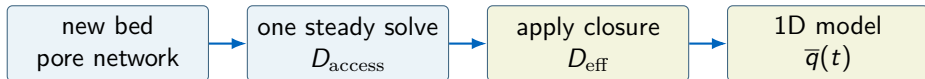
The ensemble tests arrangement effects under otherwise identical conditions.

Development and later use are two different workflows

Develop closure



Predict for a new bed



For a new bed, the transient pore-network adsorption model is no longer required.

Pore-network geometry and throat diffusion

Network construction

- four bead centres define a Delaunay tetrahedron; its circumcentre is a pore body
- tetrahedra sharing a triangular face are joined by a throat
- topology comes from particle coordinates, not voxels

A low-discrepancy Sobol sample of 2^{22} points allocates the analytically calculated void volume to pore bodies; it does not define connectivity.

Diffusion through throat $i-j$

$$G_{ij} = D_m \frac{A_{ij}}{L_{ij}}, \quad \dot{n}_{ij} = G_{ij}(c_j - c_i).$$

$L_{ij} = \|\mathbf{x}_j - \mathbf{x}_i\|$ is the pore-centre distance. The irregular opening between three beads is approximated by an equivalent circular area

$$A_{ij} = \pi r_{ij}^2,$$

where r_{ij} is the clearance of the shared Delaunay face.

The circular throat area is a geometric modelling assumption; G_{ij} has units of $\text{m}^3 \text{s}^{-1}$.

Langmuir equilibrium and bead uptake

Local equilibrium loading

$$p_k = c_k RT, \quad q_k^* = \frac{q_s b p_k}{1 + b p_k}.$$

Finite-rate uptake

$$\frac{dq_k}{dt} = k_{\text{LDF}} (q_k^* - q_k).$$

q_s is the maximum CO₂ loading [mol/kg adsorbent]. b is the Langmuir affinity, p_k is the local CO₂ partial pressure, and q_k^* is the corresponding equilibrium loading. The product $b p_k$ is dimensionless.

Meaning of the rate equation

- q_k is the current loading of bead k
- k_{LDF} controls how quickly q_k approaches q_k^*
- gas removed from adjacent pores is added to the bead

Pore-network outputs

- pore concentrations $c_i(t)$
- particle loadings $q_k(t)$
- mean uptake $\bar{q}_{\text{PNM}}(t)$

The coupled diffusion–adsorption calculation conserves the total amount of CO₂.

The homogeneous 1D uptake model

Axial diffusion and adsorption

$$\varepsilon_b \frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial z^2} - (1 - \varepsilon_b) \rho_p \frac{\partial q}{\partial t},$$

$$\frac{\partial q}{\partial t} = k_{\text{LDF}} [q^*(c) - q].$$

For each trial D , solve both equations together for $c(z, t)$ and $q(z, t)$ in 100 cells, with prescribed initial and boundary conditions.

Quantity compared with the network

$$\bar{q}_{1\text{D}}(t; D) = \frac{1}{H} \int_0^H q(z, t; D) dz.$$

The 1D model uses the same bed height, porosity, inlet condition, Langmuir parameters, LDF coefficient and initial state as the pore network. Only D changes during fitting.

For every trial D , the 1D solver produces a complete mean uptake curve.

D_{access} and D_{eff} answer different questions

D_{access} : geometric accessibility

- fix concentration at inlet and interior plane
- solve one steady, linear diffusion problem
- no adsorption and no time integration

$$D_{\text{access}} = \frac{G_{\text{access}} H}{A_{\text{tube}}}.$$

G_{access} is total steady flux per concentration difference. H is bed height; A_{tube} is tube area.

D_{eff} : transient reduced-model coefficient

- full transient network gives $\bar{q}_{\text{PNM}}(t)$
- repeated 1D solves give $\bar{q}_{1\text{D}}(t; D)$

Define the curve residual

$$e_m(D) = \bar{q}_{1\text{D}}(t_m; D) - \bar{q}_{\text{PNM}}(t_m).$$

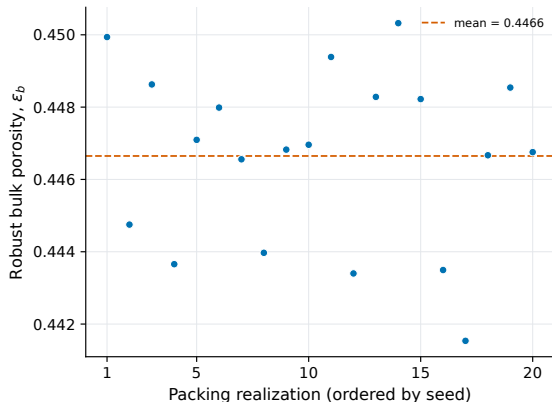
$$\text{NRMSE}(D) = \frac{\sqrt{\sum_m w_m e_m(D)^2}}{\bar{q}_{\text{PNM}}(t_f)},$$

$$D_{\text{eff}} = \arg \min_D \text{NRMSE}(D).$$

The trapezoidal weights w_m account for unequal time spacing and satisfy $\sum_m w_m = 1$.

D_{access} is cheap because it avoids adsorption dynamics and optimization.

The twenty beds have nearly equal bulk porosity



Bulk-porosity CV: 0.55%.

Controlled ensemble

- same bead count and diameter
- same tube geometry
- only the insertion seed changes

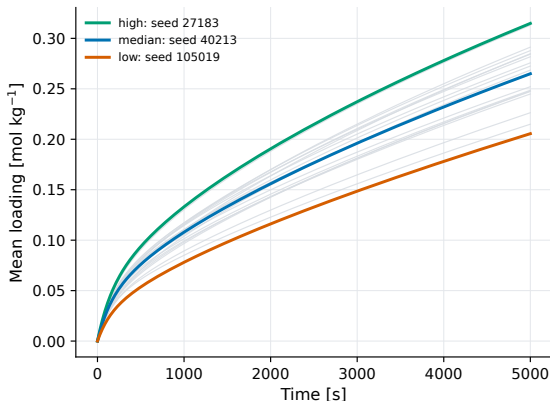
CV: coefficient of variation

$$CV_{\epsilon} = s_{\epsilon} / \bar{\epsilon}$$

s_{ϵ} : standard deviation

$\bar{\epsilon}$: mean porosity

Similar porosity still produces different uptake curves

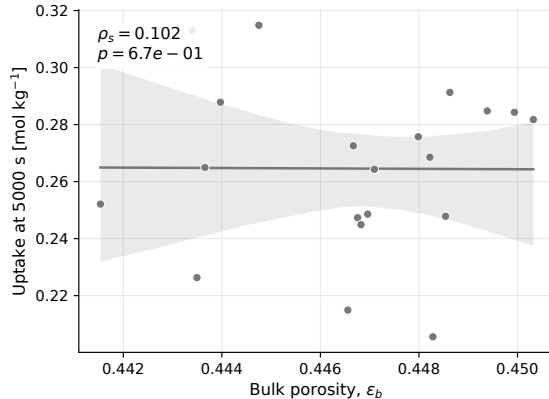


Network reference

- all twenty uptake curves
- highlighted beds span low, median and high final uptake
- identical adsorption parameters

At 5000 s: 0.206–0.315 mol kg⁻¹, with an uptake CV of 11.0%.

Bulk porosity alone does not explain final uptake

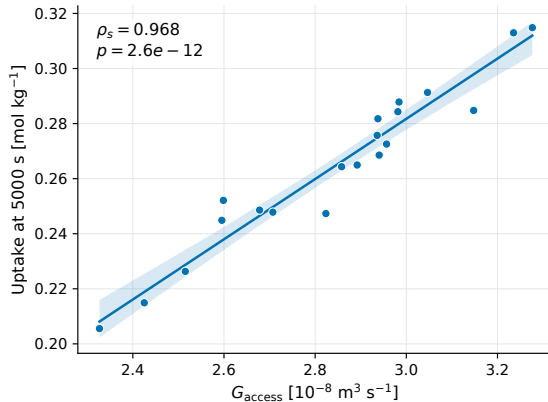


Porosity baseline

- each point represents one bed
- weak rank association
- wide uncertainty around the fitted relationship

Void volume alone does not describe access to the adsorbent.

Accessibility explains differences in final uptake



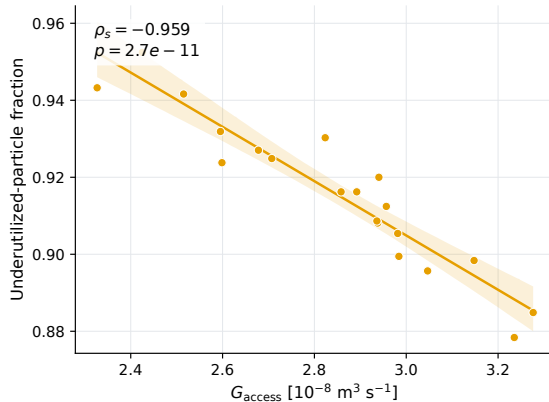
Steady network descriptor

- G_{access} measures inlet-to-interior conductance
- greater conductance accompanies greater uptake

Spearman $\rho_s = 0.968$.

Inlet-connected pathways explain variability that porosity misses.

Greater accessibility reduces particle underutilization



Higher G_{access} accompanies fewer underutilized particles.

Underutilized fraction

$$f_{\text{under}} = \frac{1}{N_p} \sum_{k=1}^{N_p} I_k.$$

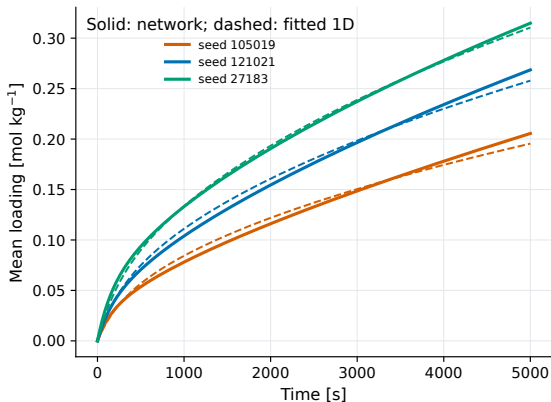
$I_k = 1$ when

$$q_k(t_f) < 0.5q^*(p_{\text{inlet}}).$$

$t_f = 5000$ s.

The indicator equals one for a particle below half the inlet-equilibrium loading.

One fitted D_{eff} reproduces each uptake curve



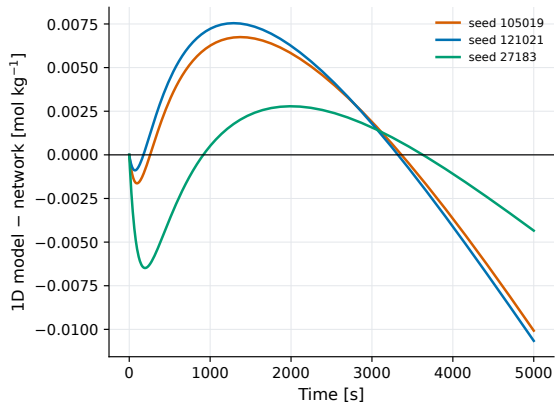
Per-bed optimization

1. solve the 1D model for a trial D
2. evaluate time-weighted NRMSE against the network curve
3. use previous errors to select the next D and repeat (optimization)

Solid: network.
Dashed: fitted 1D model.

Across twenty beds, the mean fitted-curve NRMSE is 1.58%.

Residuals show the remaining reduction error



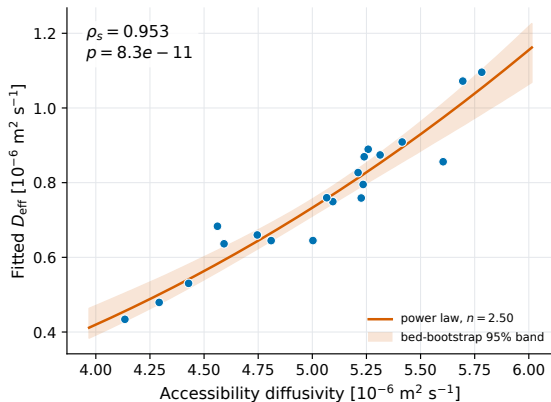
Curve residual

$$e(t) = \bar{q}_{1D}(t) - \bar{q}_{PNM}(t).$$

- positive: excess predicted uptake
- negative: insufficient predicted uptake

Time-weighted NRMSE across all beds: 0.84–2.46%.

A positive power law relates accessibility to diffusivity



Selected closure

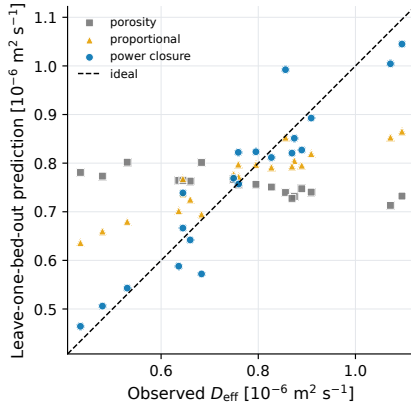
$$\frac{D_{\text{eff}}}{D_m} = C \left(\frac{D_{\text{access}}}{D_m} \right)^n.$$

$$D_{\text{access}} = \frac{G_{\text{access}} H}{A_{\text{tube}}}.$$

The shaded band shows bootstrap uncertainty in the fitted relationship.

The fitted exponent is $n \approx 2.50$ for this ensemble.

The closure predicts diffusivity for a withheld bed



Closures compared

$$D_{\text{eff}} = a_0 + a_1 \varepsilon_b,$$

$$D_{\text{eff}} = \alpha D_{\text{access}},$$

$$\frac{D_{\text{eff}}}{D_m} = C \left(\frac{D_{\text{access}}}{D_m} \right)^n.$$

Fit on nineteen beds, then predict the remaining bed.

LOOCV R^2 : power law 0.890, proportional 0.555, porosity -0.234.

Why not predict uptake directly from G_{access} ?

Direct endpoint relation

$$G_{\text{access}} \longrightarrow \bar{q}(5000 \text{ s})$$

- strong correlation: $\rho_s = 0.968$
- ranks beds or predicts one endpoint
- valid at the selected time and conditions

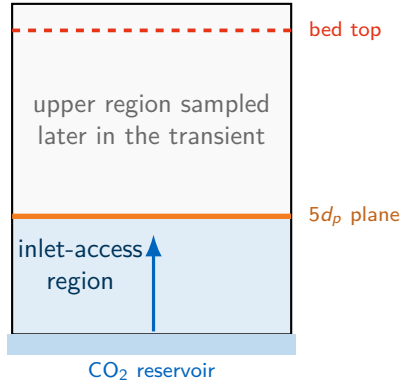
Transport closure

$$G_{\text{access}} \longrightarrow D_{\text{access}} \longrightarrow D_{\text{eff}} \longrightarrow \bar{q}(t)$$

- generates the complete uptake curve
- retains conservative 1D transport
- gives axial $c(z, t)$ and $q(z, t)$ profiles

If only uptake at 5000 s is needed, G_{access} is sufficient; D_{eff} is needed for a reusable transient model.

Why use an interior plane?



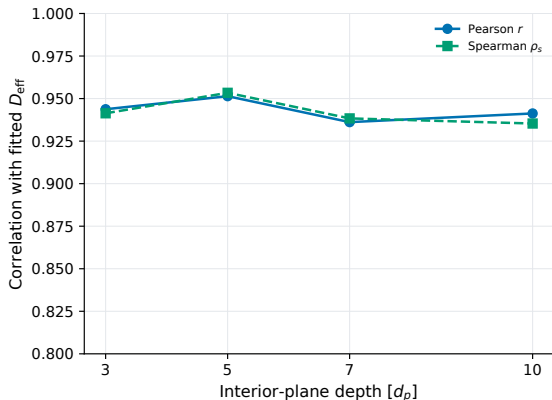
A plane at $5d_p$ includes

- the inlet openings
- second-layer and deeper bottlenecks
- parallel paths connecting inlet to the plane

The bed top answers a different question

- measure conductance through the entire bed
- include upper bottlenecks not sampled early

Accessibility remains correlated across diagnostic depths



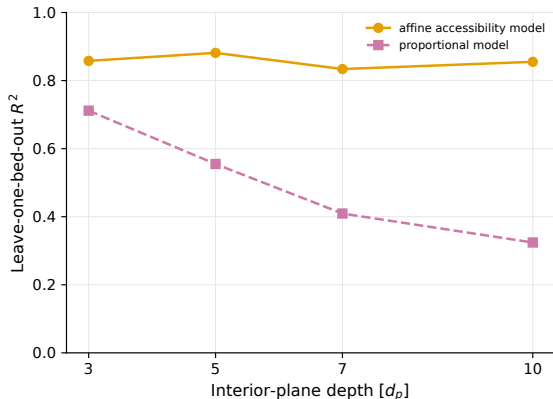
Depth sensitivity

- planes at 3, 5, 7 and $10d_p$
- Pearson: linear association
- Spearman: rank association

Depth is measured above each network's lowest pore.

Both correlations remain high across the tested depths.

Predictive performance is stable across diagnostic depths



Models in this comparison

- affine accessibility model
- proportional model

The $5d_p$ plane gives the highest affine-model score in this ensemble.

The affine accessibility model remains predictive from $3d_p$ to $10d_p$.

What the present evidence establishes

Supported

- porosity alone misses important packing variability
- inlet-connected topology controls transient utilization
- one D_{eff} reproduces each detailed curve accurately
- D_{access} provides an interpretable predictive closure

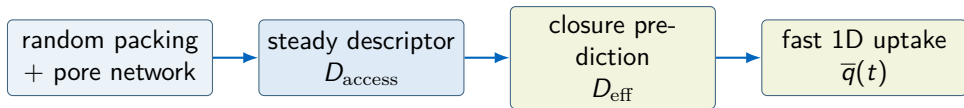
Not yet claimed

- industrial breakthrough performance
- effects of flow, heat or water competition
- transfer to other bead sizes or D_{tube}/d_p
- a universal power-law exponent

The contribution is a structure-to-model closure for a controlled diffusion-driven adsorption problem.

Pore accessibility retains what bulk porosity loses

Equal void volume does not imply equal transport. How that void space connects to the inlet matters.



20 beds
controlled ensemble

1.58%
mean reduction error

$R^2 = 0.890$
withheld-bed closure